



Field and Wave Electromagnetics

电磁场与电磁波

Chapter 7: Time-Varying Fields and Maxwell's Equations

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Chapter Title		课时	
1	Static Electric Fields	3	静电场： (1) 期末<15%，不考大题； (2) 磁矢等难度大尽量简化
2	Solution of Electrostatic Problems		
4	Steady Electric Currents	2	
5	Basic Static Magnetic Fields	2	
7	Time-varying Fields and Maxwell's Equations	5	
9	Plane Electromagnetic Waves	11	电压计算等适当简化， 重点强化章节，期末有一道 推导大题
13	Waveguides and Cavity Resonators	11	
16	Antennas and Radiating systems	3	
机动/答疑		2	
		40	

1 总体原则：弱化电路相关(V,A,I,U)，强化Photonics相关(E,H)，突出本课程对光电专业支撑作用

2 删掉“传输线”，增加“天线”；删除“镜像法”（分离变量法保留）

3 课时可以根据情况适当调整，预留2课时为机动/答疑课时；

4 平时成绩40分，（作业上交10分+点名10分，作业完成情况20分）

5 作业完成情况20分学生两种选择（默认选择a）：

（a）作业上交点名20分+作业完成情况(绑定期末作业题)20分

（b）助教随机抽出4个作业大题，每题5分，1小时完成(答疑课时候完成)，得分视为平时成绩。

Chapter 3: Static Electric Fields

Field: A spatial distribution of a scalar or vector quantity which may or may not be a function of time.

1. Fundamental postulates of electrostatic in free space

2. Coulomb's law

3. Gauss's law

4. Conductors in static electric field

5. Dielectrics in static electric field

6. Electric flux density

7. Boundary conditions for electrostatic fields

8. Capacitance and capacitors

9. Electrostatic energy and forces

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \oint_S \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$\nabla \times \vec{E} = 0 \quad \oint_C \vec{E} \cdot d\vec{l} = 0$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\oint_S \vec{D} \cdot d\vec{s} = Q$$

Chapter 4: Solutions of Electrostatic Problems

4-1 Introduction

4-2 Poisson's and Laplace's Equations $\nabla^2 V = -\frac{\rho}{\epsilon}$ $\nabla^2 V = 0$

4-3 Uniqueness of Electrostatic Solutions

4-4 Methods of Images

4-5 Boundary-Value Problems in Cartesian Coordinates

4-6 Boundary-Value Problems in Cylindrical Coordinates

4-7 Boundary-Value Problems in Spherical Coordinates

Chapter 5: Steady Electric Currents

5-1 Introduction

5-2 Current density and Ohm's law

5-3 Electromotive force and Kirchhoff's voltage law

5-4 Continuity of equation and Kirchhoff's current law

5-5 Power dissipation and Joule's law

5-6 Boundary conditions for current density

5-7 Resistance calculation

Chapter 6: Static magnetic fields

6-1 Introduction

6-2 Fundamental postulates of magnetostatics in free space

6-3 Vector magnetic potential

6-4 The Biot-Savart law and applications

6-5 The magnetic dipole

6-6 Magnetization and equivalent current densities

6-7 Magnetic field intensity and relative permeability

6-8 Magnetic circuits

6-9 Behavior of magnetic materials

6-10 Boundary conditions for magnetostatic fields

6-11 Inductances and inductors

6-12 Magnetic energy

6-13 Magnetic forces and torques

- Static Electric Field in
free space/dielectric
(differential/integral form)

- Static Magnetic Field in
free space/dielectric
(differential/integral form)

- Steady Current density
(differential/integral form)

- **Static** Electric Field in
free space/dielectric
(**differential/integral** form)

$$\begin{aligned}\nabla \cdot \vec{E} &= \frac{\rho}{\epsilon_0} & \oint_S \vec{E} \cdot d\vec{s} &= \frac{Q}{\epsilon_0} \\ \nabla \times \vec{E} &= 0 & \oint_C \vec{E} \cdot d\vec{l} &= 0 \\ \nabla \cdot \vec{D} &= \rho & \oint_S \vec{D} \cdot d\vec{s} &= Q\end{aligned}$$

- **Steady** Current density
(**differential/integral** form)

- **Static** Magnetic Field in
free space/dielectric
(**differential/integral** form)

$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$
$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$	$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I$

$\nabla \cdot \mathbf{J} = 0$	$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0$
$\nabla \times \left(\frac{\mathbf{J}}{\sigma} \right) = 0$	$\oint_C \frac{1}{\sigma} \mathbf{J} \cdot d\boldsymbol{\ell} = 0$

Electrostatics ↔ Magnetostatics	
E	B
D	H
ϵ	$\frac{1}{\mu}$
P	-M
ρ	J
V	A
$\cdot$	x
x	$\cdot$

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}$$

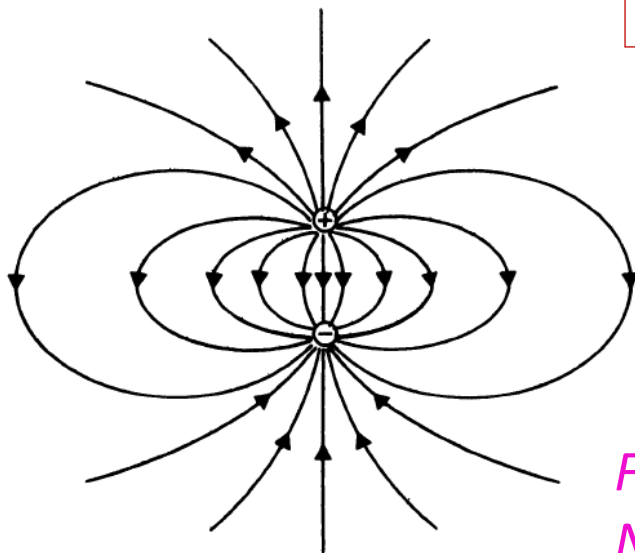
Electric and magnetic dipoles

Electric dipole

$$\mathbf{p} = q\mathbf{d}.$$

$$V = \frac{\mathbf{p} \cdot \mathbf{a}_R}{4\pi\epsilon_0 R^2}$$

$$\mathbf{E} = \frac{p}{4\pi\epsilon_0 R^3} (\mathbf{a}_R 2 \cos \theta + \mathbf{a}_\theta \sin \theta).$$



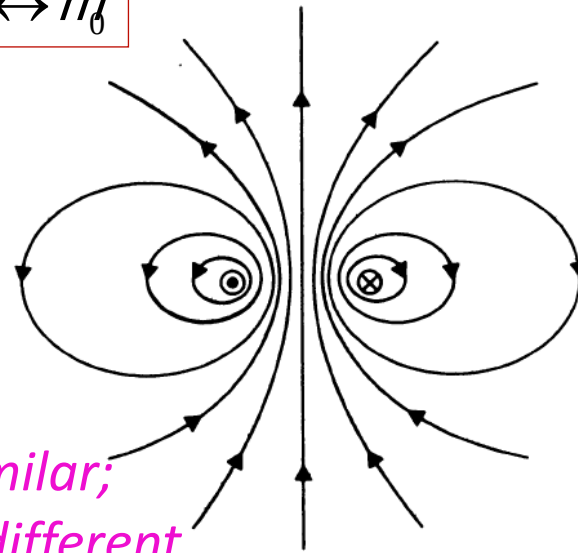
$$p \leftrightarrow m, \quad e_0^{-1} \leftrightarrow m_0$$

magnetic dipole

$$\mathbf{m} = \mathbf{a}_z I \pi b^2 = \mathbf{a}_z I S = \mathbf{a}_z m$$

$$\mathbf{A} = \frac{\mu_0 \mathbf{m} \times \mathbf{a}_R}{4\pi R^2}$$

$$\mathbf{B} = \frac{\mu_0 m}{4\pi R^3} (\mathbf{a}_R 2 \cos \theta + \mathbf{a}_\theta \sin \theta)$$



*Far-field is similar;
Near-field is different*

Chapter 7: Time-Varying Fields and Maxwell's Equations

7-1 Introduction

7-2 Faraday's Law of Electromagnetic Induction

7-3 Maxwell's Equations

7-4 Potential Functions

7-5 Electromagnetic Boundary Equations

7-6 Wave Equations and Their Solutions

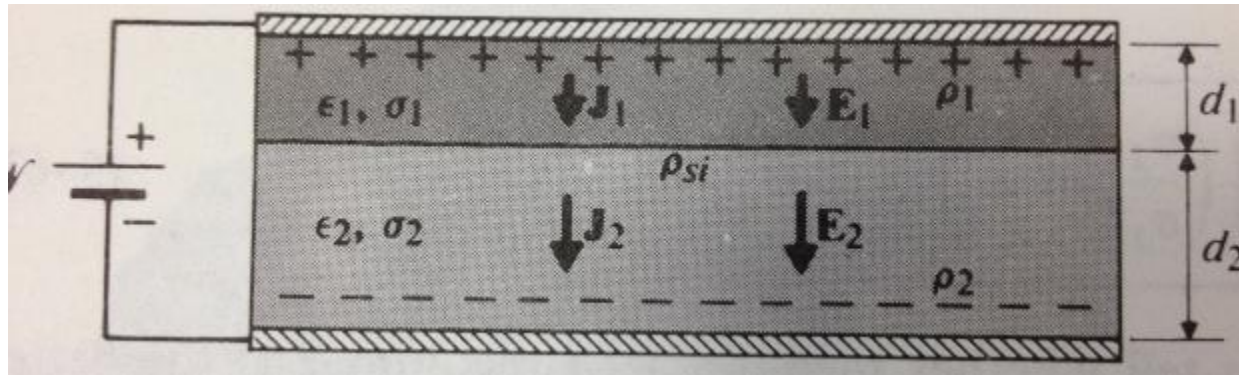
7-7 Time Harmonics Fields

7-1 Introduction

Fundamental relations for electrostatic and magnetostatic models

Fundamental Relations	Electrostatic Model	Magnetostatic Model
Governing equations	$\nabla \times \mathbf{E} = 0$ $\nabla \cdot \mathbf{D} = \rho$	$\nabla \cdot \mathbf{B} = 0$ $\nabla \times \mathbf{H} = \mathbf{J}$
Constitutive relations (linear and isotropic media)	$\mathbf{D} = \epsilon \mathbf{E}$	$\mathbf{H} = \frac{1}{\mu} \mathbf{B}$

- In the **static** case (**non-time-varying**), electric vectors $\mathbf{E}$ and $\mathbf{D}$ are independent with magnetic vectors $\mathbf{B}$ and $\mathbf{H}$.
- In a conductive medium, electric and magnetic fields may both exist and form an **electromagnetostatic** field.



- In a conductive medium, electric and magnetic fields may both exist and form an ***electromagnetostatic*** field.

$$\mathbf{J} = \sigma \mathbf{E} \quad (\text{A/m}^2),$$

Maxwell's First equation

Faraday's Law (1831): the quantitative relationship between the induced emf and the rate of change of magnetic flux linking a conducting loop.

Fundamental postulate for Electromagnetic Induction

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

- It applies to every point in the space.
- $\mathbf{E}$ is **nonconservative** in a **time-varying** magnetic flux density $\mathbf{B}$.

The integral form over a surface (applying Stokes's theorem):

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

$$\frac{\partial \vec{B}}{\partial t} = \frac{d\vec{B}}{dt} - \frac{\partial \vec{B}}{\partial x} \times \frac{\partial \vec{x}}{\partial t} \quad \vec{B} = \vec{B}(\vec{x}, t)$$

■ **General form of Faraday's law** for a moving circuit in a time varying magnetic field

$$\oint_C \mathbf{E}' \cdot d\boldsymbol{\ell} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} + \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\boldsymbol{\ell} \quad (7-34)$$

Total emf induced in the moving frame

Transformer emf

Motional emf

感应电动势包含两种：感生电动势 + 动生电动势

7-3 Maxwell's Equations

A time-varying magnetic field induces an electric :

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

$$\nabla \times \mathbf{H} = \mathbf{J},$$

$$\nabla \cdot \mathbf{D} = \rho,$$

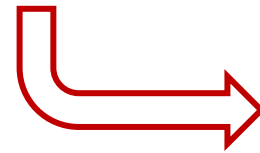
$$\nabla \cdot \mathbf{B} = 0.$$

the principle of conservation of charge

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}.$$

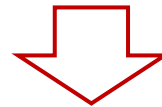
$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J}$$

In a time varying case, $\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$.



In general, not true

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t}$$



$$\nabla \cdot \mathbf{D} = \rho,$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$



- Time-varying electric field will generate magnetic field
- $\partial \mathbf{D} / \partial t$ is necessary for the consistence with the principle of conservation of charge

$\partial \mathbf{D} / \partial t$: **Displacement current density** **SI: A/m²**

- One of major contributions of Jame Clerk Maxwell (1831-1879)

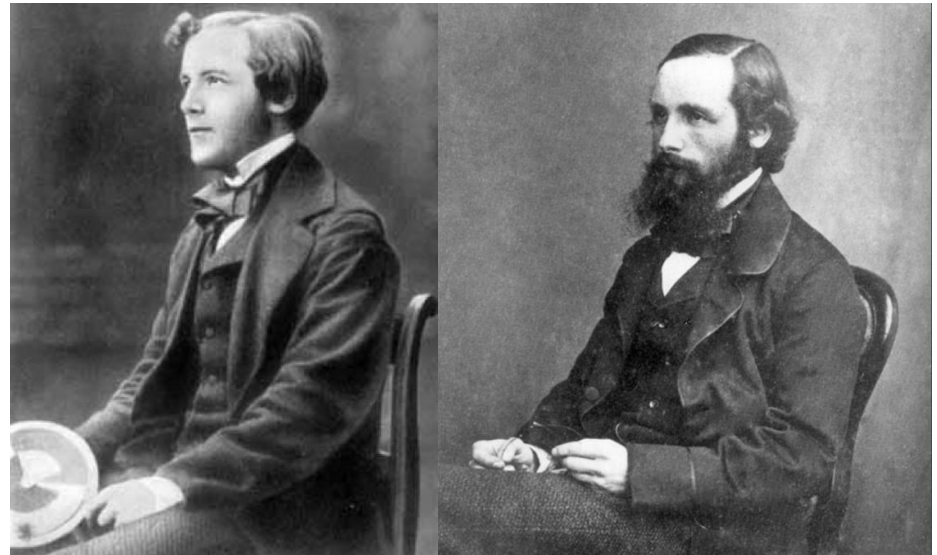
Maxwell's Equations

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \cdot \vec{B} = 0$$



James Clerk Maxwell, 1831 ~ 1879
British, Physicist and mathematician

J: density of free current (= convection current + conduction current)

ρ: volume density of free charge

■ Maxwell's equations

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t},$$

$$\nabla \cdot \mathbf{D} = \rho,$$

$$\nabla \cdot \mathbf{B} = 0.$$

■ Charge conservation equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}.$$

■ Lorentz equation

$$\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

All macroscopic electromagnetic phenomena could be explained by the above three equations.

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \cdot \vec{B} = 0$$

- Not all independent
- 12 unknowns: ***E, D, B, H***
- *12 scalar equations*
- **Constituent relations**

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\mathbf{H} = \mathbf{B}/\mu,$$

7-3-1 Integral form of Maxwell's equations

Take the surface integral of both sides of the curl equations over an open surface S with a contour C and apply Stokes's theorem.

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \Rightarrow \quad \oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \quad \Rightarrow \quad \oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{s}.$$

Taking the volume integral of both sides of the divergence equations over a volume V with a closed surface S and using divergence theorem

$$\nabla \cdot \mathbf{D} = \rho, \quad \Rightarrow \quad \oint_S \mathbf{D} \cdot d\mathbf{s} = \int_V \rho \, dv$$

$$\nabla \cdot \mathbf{B} = 0. \quad \Rightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \quad (\text{no isolated magnetic charge})$$

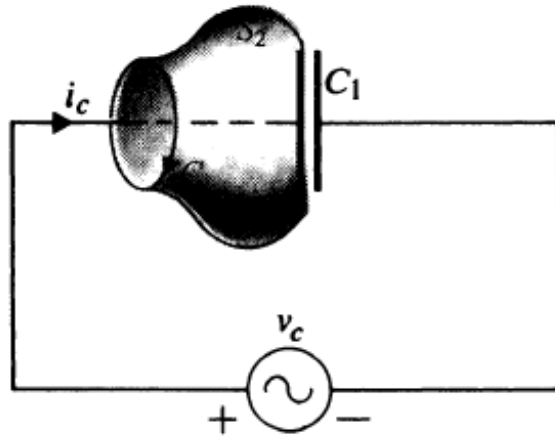
Maxwell's equations

Differential Form	Integral Form	Significance
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}$	Faraday's law
$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}$	Ampère's circuital law
$\nabla \cdot \mathbf{D} = \rho$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$	Gauss's law
$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$	No isolated magnetic charge

7-5 An a-c voltage source of amplitude V_0 and angular frequency ω , $v_c = V_0 \sin \omega t$, is connected across a parallel-plate capacitor C_1 .

(a) Verify that the displacement current in the capacitor is the same as the conduction current in the wires.

(b) Determine the magnetic field intensity H at a distance r from the wire.



(a) The conduction current: $i_c = C_1 \frac{dv_c}{dt} = C_1 V_0 \omega \cos \omega t$ (A) $C_1 = \epsilon \frac{A}{d}$

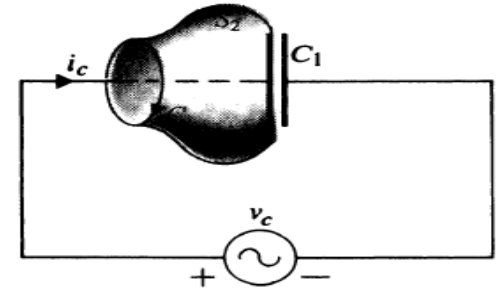
With a voltage v_c appearing between the plates, $E = v_c/d$

$$D = \epsilon E = \epsilon \frac{V_0}{d} \sin \omega t.$$

The displacement current:

$$i_D = \oint_A \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S} = \left(\epsilon \frac{A}{d} \right) V_0 \omega \cos \omega t = C_1 V_0 \omega \cos \omega t = i_c.$$

(b) The magnetic field intensity at a distance r from the conducting : Ampere's circuital law to contour C . Two typical open surfaces with rim C may be chosen :



(1) A planar disk surface S_1 , or (2) a curved surface S_2 passing through the dielectric medium.

$$\oint_C \vec{H} \times d\vec{\ell} = \oint_{S_1} \vec{J} \cdot d\vec{s} + \frac{1}{\mu_0} \frac{d}{dt} \oint_{S_2} \vec{D} \cdot d\vec{s}$$

For the surface S_1 , only the first term on the right side is nonzero.

$$\oint_{S_1} \vec{J} \cdot d\vec{s} = i_c = C_1 V_0 \omega \cos \omega t.$$

$$\oint_C \vec{H} \times d\vec{\ell} = 2\pi r H_f$$

$$H_f = \frac{C_1 V_0}{2\pi r} \omega \cos \omega t \quad (A/m)$$

Since the surface S_2 passes through the dielectric medium, no conduction current flows through S_2 . As we have shown in part(a), $i_D = i_C$. Hence we obtain the same result whether surface S_1 or S_2 is chosen.

7-4 Potential Functions

Vector magnetic potential $\mathbf{A}$:

$$\mathbf{B} = \nabla \times \mathbf{A} \quad \leftarrow \quad \nabla \cdot \mathbf{B} = 0$$

Substitute the above equation into Faradays' law

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t} (\nabla \times \mathbf{A})$$

or

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0.$$



$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla V,$$

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \quad (\text{V/m}).$$

- In the static case, $\partial \mathbf{A} / \partial t = 0$, $\mathbf{E} = -\nabla V$.
- For time-varying fields, $\mathbf{E}$ depends on both V and $\mathbf{A}$.
- $\mathbf{E}$ and $\mathbf{B}$ are coupled.

In the static or quasi-static state we have the solutions of Poisson's equations

$$V = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\rho}{R} dv', \quad \mathbf{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\mathbf{J}}{R} dv'.$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\mathbf{B} = \nabla \times \mathbf{A}$$

$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla V,$$

$$\nabla \times \nabla \times \mathbf{A} = \mu \mathbf{J} + \mu \epsilon \frac{\partial}{\partial t} \left(-\nabla V - \frac{\partial \mathbf{A}}{\partial t} \right),$$



$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu \mathbf{J} - \nabla \left(\mu \epsilon \frac{\partial V}{\partial t} \right) - \mu \epsilon \frac{\partial^2 \mathbf{A}}{\partial t^2}$$

or

$$\nabla^2 \mathbf{A} - \mu \epsilon \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu \mathbf{J} + \nabla \left(\nabla \cdot \mathbf{A} + \mu \epsilon \frac{\partial V}{\partial t} \right).$$

We take the liberty to choose the divergence of $\mathbf{A}$

$$\nabla \cdot \mathbf{A} + \mu\epsilon \frac{\partial V}{\partial t} = 0,$$

- *Lorentz conditions or Lorentz gauge for the potentials*
- *Consistent with **the equation of continuity** (Problem P7-12)*
- *Reduces to $\nabla \cdot \mathbf{A} = 0$ for the static field*

$$\nabla^2 \mathbf{A} - \mu\epsilon \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu\mathbf{J}.$$

Nonhomogeneous wave equations for vector potential $\mathbf{A}$

A corresponding equation for V can be obtained

$$-\nabla \cdot \epsilon \left(\nabla V + \frac{\partial \mathbf{A}}{\partial t} \right) = \rho,$$

$$\nabla^2 V - \mu\epsilon \frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\epsilon},$$

Nonhomogeneous wave equations for scalar potential V

Lorentz condition $\nabla \cdot \mathbf{A} + \mu\epsilon \frac{\partial V}{\partial t} = 0$, *uncouples wave equations for $\mathbf{A}$ and V .*

$$\nabla^2 \mathbf{A} - \mu\epsilon \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu\mathbf{J}.$$

7-5 Electromagnetic Boundary Conditions

■ Boundary condition for tangential $\mathbf{E}$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \longleftrightarrow \oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt} \Rightarrow E_{1t} = E_{2t}$$

1. *The tangential component of an $\mathbf{E}$ field is continuous across an interface.*

■ Boundary condition for tangential $\mathbf{H}$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \longleftrightarrow \oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s} \Rightarrow \mathbf{a}_{n2} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$$

2. *The tangential component of an $\mathbf{H}$ field is discontinuous across an interface where a surface current exists, the amount of discontinuity being determined by Eq. (7-66b).*

■ Boundary condition for normal $\mathbf{D}$

$$\nabla \cdot \mathbf{D} = \rho \quad \longleftrightarrow \quad \oint_S \mathbf{D} \cdot d\mathbf{s} = Q \quad \longrightarrow \quad \boxed{\mathbf{a}_{n2} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s}$$

3. *The normal component of a $\mathbf{D}$ field is discontinuous across an interface where a surface charge exists, the amount of discontinuity being determined by Eq. (7-66c).*

■ Boundary condition for normal $\mathbf{B}$

$$\nabla \cdot \mathbf{B} = 0 \quad \longleftrightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{s} = 0 \quad \longrightarrow \quad \boxed{B_{1n} = B_{2n}}$$

4. *The normal component of a $\mathbf{B}$ field is continuous across an interface.*

Interface between two lossless linear media

$$\rho_s = 0 \text{ and } \mathbf{J}_s = 0$$

$$E_{1t} = E_{2t} \rightarrow \frac{D_{1t}}{D_{2t}} = \frac{\epsilon_1}{\epsilon_2}$$

$$H_{1t} = H_{2t} \rightarrow \frac{B_{1t}}{B_{2t}} = \frac{\mu_1}{\mu_2}$$

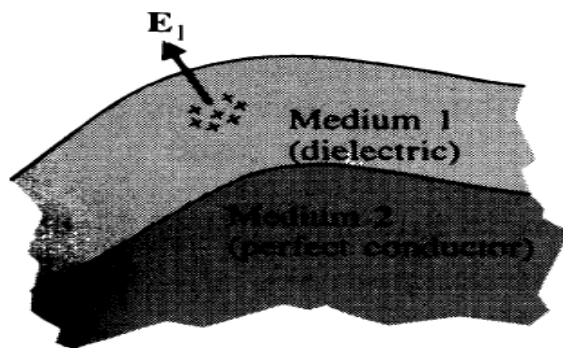
$$D_{1n} = D_{2n} \rightarrow \epsilon_1 E_{1n} = \epsilon_2 E_{2n}$$

$$B_{1n} = B_{2n} \rightarrow \mu_1 H_{1n} = \mu_2 H_{2n}$$

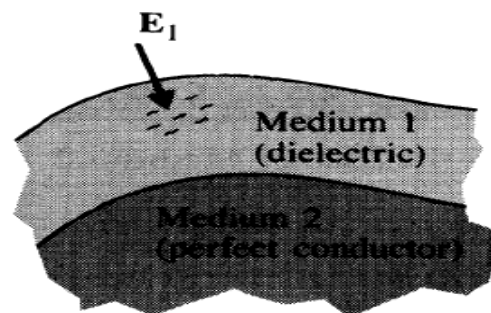
Interface between a dielectric and a perfect conductor

Boundary Conditions between a Dielectric (Medium 1) and a Perfect Conductor (Medium 2) (Time-Varying Case)

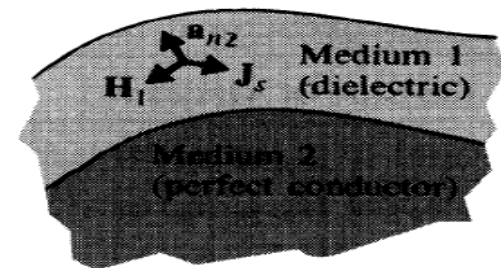
On the Side of Medium 1	On the Side of Medium 2
$E_{1t} = 0$	$E_{2t} = 0$
$\mathbf{a}_{n2} \times \mathbf{H}_1 = \mathbf{J}_s$	$H_{2t} = 0$
$\mathbf{a}_{n2} \cdot \mathbf{D}_1 = \rho_s$	$D_{2n} = 0$
$B_{1n} = 0$	$B_{2n} = 0$



(a)



(b)



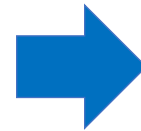
(c)

7-6 Wave Equations and their Solutions

For given charge ρ and current distribution $\mathbf{J}$, first find $\mathbf{A}$ and V , and then calculate $\mathbf{E}$ and $\mathbf{B}$.

$$\nabla^2 \mathbf{A} - \mu\epsilon \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu \mathbf{J}.$$

$$\nabla^2 V - \mu\epsilon \frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\epsilon},$$



$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$$

$$\mathbf{B} = \nabla \times \mathbf{A}$$

Source of $\mathbf{E}$: (1) charges (static or varying) ; (2) varying magnetic field.

Source of $\mathbf{B}$: electric current (free electron or varying polarized charges).

1. Maxwell's equations

Differential Form	Integral Form	Significance
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}$	Faraday's law
$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}$	Ampère's circuital law
$\nabla \cdot \mathbf{D} = \rho$	$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q$	Gauss's law
$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{s} = 0$	No isolated magnetic charge

2. Electromagnetic Boundary Conditions

3. Calculate **E** and **B** with **A** and **V**.

4. the principle of conservation of charge/Continuity of equation

5. Ohm's law and Joule's law (microscopic form)

6. Electromagnetic force (Lorentz equation)

7-6-1 Solutions of Wave Equations for Potentials

- Assume an point charge at time t , $\rho(t)\Delta v'$, at the origin of a spherical coordinate and out of the source region we have

$$\frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) - \mu\epsilon \frac{\partial^2 V}{\partial t^2} = 0. \quad \leftarrow \quad \boxed{\nabla^2 V - \mu\epsilon \frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\epsilon}},$$



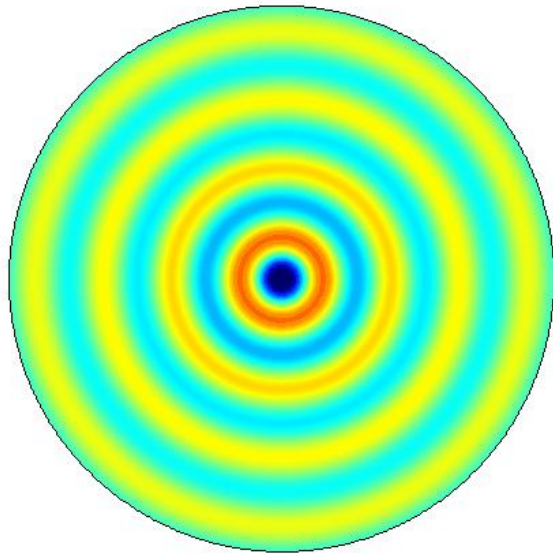
Introduce a new variable

$$V(R, t) = \frac{1}{R} U(R, t),$$

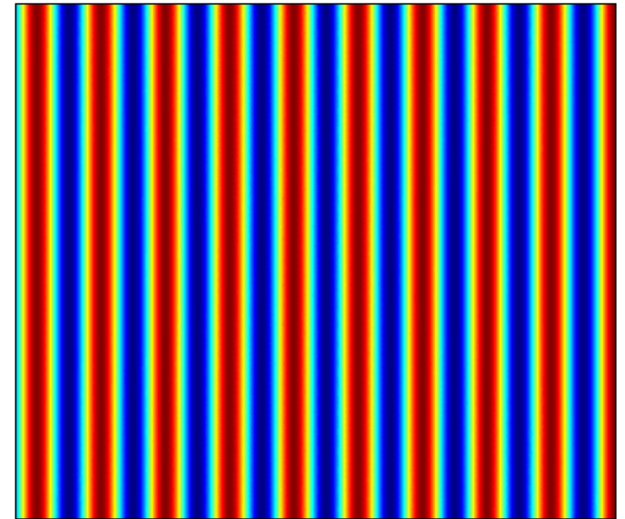
$$\frac{\partial^2 U}{\partial R^2} - \mu\epsilon \frac{\partial^2 U}{\partial t^2} = 0. \quad (7-73)$$

One-dimensional homogeneous wave equation

$V(R, t)$



$U(R, t)$



$$V^*R=U$$

Any twice differential function of $(t - R\sqrt{\mu\epsilon})$ or of $(t + R\sqrt{\mu\epsilon})$ is a solution of Eq. (7-73). But here we only select a function of $(t - R\sqrt{\mu\epsilon})$ for causality. Hence we have

$$U(R, t) = f(t - R\sqrt{\mu\epsilon}). \quad \longrightarrow \quad \boxed{\frac{\partial^2 U}{\partial R^2} - \mu\epsilon \frac{\partial^2 U}{\partial t^2} = 0.} \quad (7-73)$$

Wave travelling in the positive R direction with a velocity $1/\sqrt{\mu\epsilon}$

As we see, the function at $R + \Delta R$ at a later time $t + \Delta t$ is

$$U(R + \Delta R, t + \Delta t) = f[t + \Delta t - (R + \Delta R)\sqrt{\mu\epsilon}] = f(t - R\sqrt{\mu\epsilon}).$$

 We cannot take "+" for causality.

Thus the function retains its form if $\Delta t = \Delta R\sqrt{\mu\epsilon} = \Delta R/u$, where $u = 1/\sqrt{\mu\epsilon}$ is the **velocity of propagation**, a characteristic of the medium. From Eq. (7-72) we get

$$V(R, t) = \frac{1}{R} f(t - R/u). \quad (7-75)$$

To determine what the specific function $f(t - R/u)$ must be, we note from Eq. (3-47) that for a static point charge $\rho(t) \Delta v'$ at the origin,

$$\Delta V(R) = \frac{\rho(t) \Delta v'}{4\pi\epsilon R}. \quad (7-76)$$

$$V(R, t) = \frac{1}{R} f(t - R/u). \quad (7-75)$$

Comparison of Eqs. (7-75) and (7-76) enables us to identify

$$\Delta f(t - R/u) = \frac{\rho(t - R/u) \Delta v'}{4\pi\epsilon}.$$

Electric potential due to a charge distribution over a volume V' :

$$V(R, t) = \frac{1}{4\pi\epsilon} \int_{V'} \frac{\rho(t - R/u)}{R} dv' \quad (\text{V}).$$

Retarded scalar potential

Equation (7-77) indicates that the scalar potential at a distance R from the source at time t depends on the value of the charge density at an *earlier* time $(t - R/u)$. It takes time R/u for the effect of ρ to be felt at distance R . For this reason, $V(R, t)$ in Eq. (7-77) is called the ***retarded scalar potential***. It is now clear that a function of $(t + R/u)$ cannot be a physically useful solution, since it would lead to the impossible situation that the effect of ρ would be felt at a distant point before it occurs at the source.

Magnetic vector potential due to a current distribution over a volume V' (could be processed in the same way):

$$\mathbf{A}(R, t) = \frac{\mu}{4\pi} \int_{V'} \frac{\mathbf{J}(t - R/u)}{R} dv' \quad (\text{Wb/m}).$$

Retarded vector potential

- The electric and magnetic fields derived from A and V will be functions of $(t - R\sqrt{\mu\epsilon})$ and therefore retarded in time.
- It takes for electromagnetic waves to travel and to be felt at a distance.

7-6-2 Source-free Wave Equations

For source-free regions ($\rho=0$ and $J=0$) in simple (linear, isotropic and homogenous) non-conducting ($\sigma=0$) medium, Maxwell's equations are reduced to

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t},$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t},$$

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t},$$

$$\nabla \cdot \mathbf{D} = \rho,$$

$$\nabla \cdot \mathbf{E} = 0,$$

$$\nabla \cdot \mathbf{B} = 0.$$

$$\nabla \cdot \mathbf{H} = 0.$$

$$\nabla \times \nabla \times \mathbf{E} = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) = -\mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}$$

$$\nabla^2 \mathbf{E} - \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0;$$

$$\downarrow u = 1/\sqrt{\mu\epsilon},$$

$$\nabla^2 \mathbf{E} - \frac{1}{u^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0.$$

Similarly,

$$\nabla^2 \mathbf{H} - \frac{1}{u^2} \frac{\partial^2 \mathbf{H}}{\partial t^2} = 0.$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad (7-79a)$$

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t}, \quad (7-79b)$$

$$\nabla \cdot \mathbf{E} = 0, \quad (7-79c)$$

$$\nabla \cdot \mathbf{H} = 0. \quad (7-79d)$$

Homogeneous vector wave equations

(could be decomposed into three one-dimensional homogeneous scalar wave equations.)

7-7 Time-Harmonic Fields(时谐场)

steady-state sinusoidal field

- Arbitrary periodic time functions can be expanded into Fourier series of harmonic sinusoidal components
- Transient nonperiodic functions can be expressed as Fourier integrals
- Sinusoidal time variations of source functions will produce sinusoidal variations of $\mathbf{E}$ and $\mathbf{H}$ with the same frequency
- Electrodynamic fields can be determined in terms of those caused by the various frequency components of an arbitrary time-varying source function.
- The principle of superposition

7-7-2 Time-Harmonic Electromagnetics

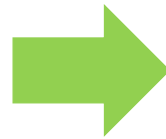
For a time-harmonic E field referring to $\cos\omega t$ can be written as

$$\mathbf{E}(x, y, z, t) = \Re e[\mathbf{E}(x, y, z)e^{j\omega t}],$$

where $\mathbf{E}(x, y, z)$ is a **vector phasor** that contains information on direction, magnitude, and phase. Phasors are, in general, complex quantities. From Eqs. (7–93), (7–87), (7–89), and (7–90) we see that, if $\mathbf{E}(x, y, z, t)$ is to be represented by the vector phasor $\mathbf{E}(x, y, z)$, then $\partial\mathbf{E}(x, y, z, t)/\partial t$ and $\int \mathbf{E}(x, y, z, t) dt$ would be represented by vector phasors $j\omega\mathbf{E}(x, y, z)$ and $\mathbf{E}(x, y, z)/j\omega$, respectively. Higher-order differentiations and integrations with respect to t would be represented by multiplications and divisions, respectively, of the phasor $\mathbf{E}(x, y, z)$ by higher powers of $j\omega$.

Time-harmonic Maxwell's equations in a simple (linear, isotropic and homogeneous) medium

$$\begin{aligned}\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{H} &= \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \\ \nabla \cdot \mathbf{D} &= \rho, \\ \nabla \cdot \mathbf{B} &= 0.\end{aligned}$$



$$\begin{aligned}\nabla \times \mathbf{E} &= -j\omega\mu\mathbf{H}, \\ \nabla \times \mathbf{H} &= \mathbf{J} + j\omega\epsilon\mathbf{E}, \\ \nabla \cdot \mathbf{E} &= \rho/\epsilon, \\ \nabla \cdot \mathbf{H} &= 0.\end{aligned}$$

- Almost exclusively deal with time-harmonic fields (and therefore with phasors)
- Phase quantities are not functions of time;
- Any quantity containing j must necessarily be a phasor.

Time-harmonic wave equations for scalar potential V and vector potential $\mathbf{A}$

non-homogenous
Helmholtz equations

$$\nabla^2 V - \mu\epsilon \frac{\partial^2 V}{\partial t^2} = -\frac{\rho}{\epsilon}, \quad \longrightarrow \quad \nabla^2 V + k^2 V = -\frac{\rho}{\epsilon}$$

$$\nabla^2 \mathbf{A} - \mu\epsilon \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu \mathbf{J}, \quad \longrightarrow \quad \nabla^2 \mathbf{A} + k^2 \mathbf{A} = -\mu \mathbf{J},$$

where wavenumber: $k = \omega \sqrt{\mu\epsilon} = \frac{\omega}{u}$

Lorentz condition

$$\nabla \cdot \mathbf{A} + \mu\epsilon \frac{\partial V}{\partial t} = 0, \quad \longrightarrow \quad \nabla \cdot \mathbf{A} + j\omega\mu\epsilon V = 0.$$

The phasor solutions (retarded) for the potential equations:

$$V(R, t) = \frac{1}{4\pi\epsilon} \int_{V'} \frac{\rho(t - R/u)}{R} dv'$$

$$V(R) = \frac{1}{4\pi\epsilon} \int_{V'} \frac{\rho e^{-jkR}}{R} dv'$$

$$\mathbf{A}(R, t) = \frac{\mu}{4\pi} \int_{V'} \frac{\mathbf{J}(t - R/u)}{R} dv'$$

$$\mathbf{A}(R) = \frac{\mu}{4\pi} \int_{V'} \frac{\mathbf{J} e^{-jkR}}{R} dv'$$

If $kR \ll 1$, equations simplify to be the static expressions

Formal procedure to determine E and H due harmonic charges and currents:

1. Find phasors $V(R)$ and $\mathbf{A}(R)$ from Eqs. (7-99) and (7-100).
2. Find phasors $\mathbf{E}(R) = -\nabla V - j\omega\mathbf{A}$ and $\mathbf{B}(R) = \nabla \times \mathbf{A}$.
3. Find instantaneous $\mathbf{E}(R, t) = \Re[\mathbf{E}(R)e^{j\omega t}]$ and $\mathbf{B}(R, t) = \Re[\mathbf{B}(R)e^{j\omega t}]$ for a cosine reference.

7-7-3 Source-free Fields in Simple Media

$$\rho = 0, \mathbf{J} = 0, \sigma = 0,$$

$$\begin{aligned}\nabla \times \mathbf{E} &= -j\omega\mu\mathbf{H}, \\ \nabla \times \mathbf{H} &= \mathbf{J} + j\omega\epsilon\mathbf{E}, \\ \nabla \cdot \mathbf{E} &= \rho/\epsilon, \\ \nabla \cdot \mathbf{H} &= 0.\end{aligned}$$

$$\begin{aligned}\nabla \times \mathbf{E} &= -j\omega\mu\mathbf{H}, \\ \nabla \times \mathbf{H} &= j\omega\epsilon\mathbf{E}, \\ \nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{H} &= 0.\end{aligned}$$

Homogenous vector
wave equations

$$\nabla^2 \mathbf{E} - \frac{1}{u^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0.$$

$$\nabla^2 \mathbf{H} - \frac{1}{u^2} \frac{\partial^2 \mathbf{H}}{\partial t^2} = 0.$$

Homogenous vector
Helmholtz's equations

$$\nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0$$

$$\nabla^2 \mathbf{H} + k^2 \mathbf{H} = 0,$$

EXAMPLE 7-7 Show that if $(\mathbf{E}, \mathbf{H})$ are solutions of source-free Maxwell's equations in a simple medium characterized by ϵ and μ , then so also are $(\mathbf{E}', \mathbf{H}')$, where

$$\mathbf{E}' = \eta \mathbf{H} \quad \text{(Principle of duality: 对偶原理)} \quad (7-107a)$$

$$\mathbf{H}' = -\frac{\mathbf{E}}{\eta}. \quad \text{PEC and PMC?} \quad (7-107b)$$

In the above equations, $\eta = \sqrt{\mu/\epsilon}$ is called the *intrinsic impedance* of the medium.

Solution We prove the statement by taking the curl and the divergence of $\mathbf{E}'$ and $\mathbf{H}'$ and using Eqs. (7-104a, b, c, and d):

$$\begin{aligned} \nabla \times \mathbf{E}' &= \eta(\nabla \times \mathbf{H}) = \eta(j\omega\epsilon\mathbf{E}) \\ &= -j\omega\epsilon\eta^2\left(-\frac{\mathbf{E}}{\eta}\right) = -j\omega\mu\mathbf{H}' \end{aligned} \quad (7-108a)$$

$$\begin{aligned} \nabla \times \mathbf{H}' &= -\frac{1}{\eta}(\nabla \times \mathbf{E}) = -\frac{1}{\eta}(-j\omega\mu\mathbf{H}) \\ &= j\omega\mu\frac{1}{\eta^2}(\eta\mathbf{H}) = j\omega\epsilon\mathbf{E}' \end{aligned} \quad (7-108b)$$

$$\nabla \cdot \mathbf{E}' = \eta(\nabla \cdot \mathbf{H}) = 0 \quad (7-108c)$$

$$\nabla \cdot \mathbf{H}' = -\frac{1}{\eta}(\nabla \cdot \mathbf{E}) = 0. \quad (7-108d)$$

Equations (7-108a, b, c, and d) are source-free Maxwell's equations in $\mathbf{E}'$ and $\mathbf{H}'$.

In a conducting medium ($\sigma \neq 0$)

$$\nabla \times \mathbf{H} = (\sigma + j\omega\epsilon)\mathbf{E} = j\omega\left(\epsilon + \frac{\sigma}{j\omega}\right)\mathbf{E} = j\omega\epsilon_c\mathbf{E}$$

Complex permittivity: $\epsilon_c = \epsilon - j\frac{\sigma}{\omega}$ (F/m).

In general case, $\epsilon_c = \epsilon' - j\epsilon''$ (F/m),

Similarly, $\mu = \mu' - j\mu''$.

-Damping loss due to the inertial property of charged particles;
-Ohmic loss in metal or semiconductor
(good conductor: $\sigma \gg \omega\epsilon$)

Complex wavenumber:

$$k_c = \omega\sqrt{\mu\epsilon_c} = \omega\sqrt{\mu(\epsilon' - j\epsilon'')}$$

Loss tangent (δ_c -loss angle):

$$\tan \delta_c = \frac{\epsilon''}{\epsilon'} \cong \frac{\sigma}{\omega\epsilon}$$

On the basis of Eq. (7–110) a medium is said to be a **good conductor** if $\sigma \gg \omega\epsilon$, and a **good insulator** if $\omega\epsilon \gg \sigma$. Thus, a material may be a good conductor at low frequencies but may have the properties of a lossy dielectric at very high frequencies. For example, a moist ground has a dielectric constant ϵ_r and a conductivity σ that are in the neighborhood of 10 and 10^{-2} (S/m), respectively. The loss tangent $\sigma/\omega\epsilon$ of the moist ground then equals 1.8×10^4 at 1 (kHz), making it a relatively good conductor. At 10 (GHz), $\sigma/\omega\epsilon$ becomes 1.8×10^{-3} , and the moist ground behaves more like an insulator.[†]

EXAMPLE 7–8 A sinusoidal electric intensity of amplitude 250 (V/m) and frequency 1 (GHz) exists in a lossy dielectric medium that has a relative permittivity of 2.5 and a loss tangent of 0.001. Find the average power dissipated in the medium per cubic meter.

Solution First we must find the effective conductivity of the lossy medium:

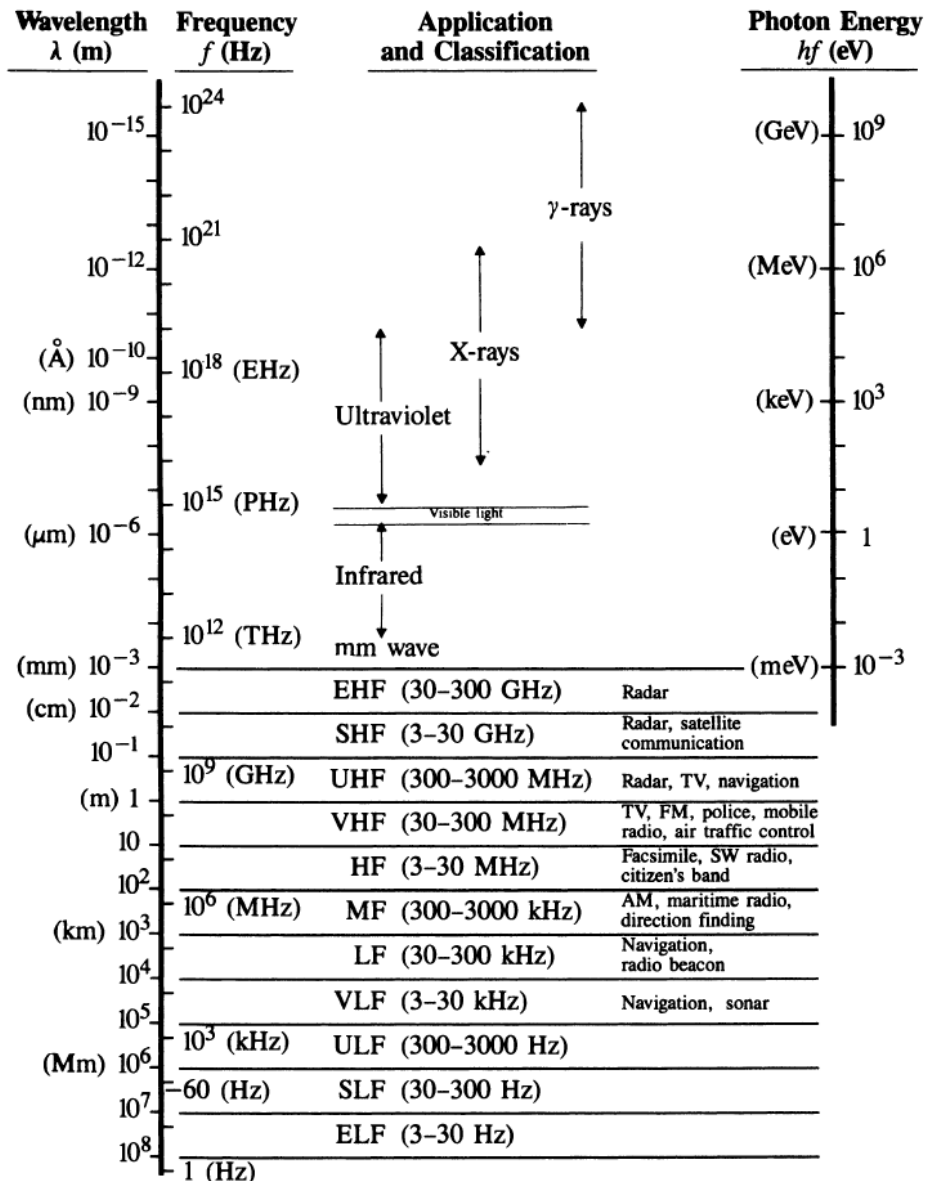
$$\begin{aligned}\tan \delta_c &= 0.001 = \frac{\sigma}{\omega \epsilon_0 \epsilon_r}, \\ \sigma &= 0.001(2\pi 10^9) \left(\frac{10^{-9}}{36\pi} \right) (2.5) \\ &= 1.39 \times 10^{-4} \text{ (S/m)}.\end{aligned}$$

The average power dissipated per unit volume is

$$\begin{aligned}p &= \frac{1}{2}JE = \frac{1}{2}\sigma E^2 \\ &= \frac{1}{2} \times (1.39 \times 10^{-4}) \times 250^2 = 4.34 \text{ (W/m}^3\text{)}.\end{aligned}$$

A microwave oven cooks food by irradiating the food with microwave power generated by a magnetron. The operating frequency is usually set at 2.45 GHz (2.45×10^9 Hz). For a beef steak that has approximately a dielectric constant of 40 and a loss tangent of 0.35 at 2.45 (GHz), calculations following those in Example 7–8 will yield $\sigma = 1.91$ (S/m) and $p = 59.6$ (kW/m³). However, since high-frequency currents in a conducting body tend to concentrate near the surface layer (due to *skin effect*—see Subsection 8–3.2), the value of p obtained here is only a rough estimate.

7-7-4 The Electromagnetic Spectrum



Band Designations for Microwave Frequency Ranges

Old†	New	Frequency Ranges (GHz)
Ka	K	26.5–40
K	K	20–26.5
K	J	18–20
Ku	J	12.4–18
X	J	10–12.4
X	I	8–10
C	H	6–8
C	G	4–6
S	F	3–4
S	E	2–3
L	D	1–2
UHF	C	0.5–1